

Twisted-Convolution Identities in Dimensions Four and Five from Shintani Ray Units

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Abstract

We prove the formal Twisted Convolution Conjecture unconditionally in dimensions four and five. Dimension four is the quadratic calibration: the principal-ghost table depends on one positive double-sine value, all 36 rank-two minors have one common quadratic factor, and Kopp’s Kronecker-limit formula together with an analytic class-number calculation evaluates that factor exactly. Dimension five is the main arithmetic case. For $K = \mathbb{Q}(\sqrt{3})$ and modulus $(5)\infty_2$, we compute all twenty-four nonzero characteristics, their positive Kopp lifts, ray classes, signs, and multipliers. Shintani’s 1978 algebraicity theorem gives an explicit safe exponent 5760 that makes his existential algebraicity effective; labeled ray-field and Frobenius certificates, rigorous Arb enclosures, and Voutier’s height bound then identify the complete reciprocal degree-eight ray packet. Exact number-field arithmetic makes all 100 rank-two minors vanish. In both dimensions complex conjugation supplies the paired shift, and integral form covariance gives the result for every admissible tuple. No Stark conjecture or **bnrstark** recognition is used as an input. A companion paper applies the same bridge to the discriminant-32 dimension-seven stratum and both dimension-eight strata; the present paper remains logically independent of those higher-dimensional certificates.

1 Overview and common convention

The principal-ghost construction of Appleby–Flammia–Kopp [3], continuing the SIC–Stark program of Kopp [4], associates convention-sensitive Shintani–Faddeev real-multiplication values to a finite Weyl reconstruction. The formal TCC asks for two shifted twisted convolutions to vanish, equivalently for the reconstructed ghost matrices to be rank-one idempotents. Our proof architecture is

characteristic-to-ray and multiplier bridge
↓
analytic identification of a labeled ray-unit packet
↓
exact vanishing of every rank-two minor
↓
both formal shifts and integral-form transport.

To remove the standard double-sine ambiguity once and for all, define

$$S_2(z \mid \omega_1, \omega_2) := \frac{\Gamma_2(z \mid \omega_1, \omega_2)}{\Gamma_2(\omega_1 + \omega_2 - z \mid \omega_1, \omega_2)}. \quad (1)$$

Thus S_2 is the reciprocal of Kopp’s $\text{Sin}_{2, \text{Kopp}}$ convention [1, Definition 4.22, equation (4.8)].

1.1 AFK terminology and the formal TCC

We recall the data used in the theorem; this also fixes our transcription of AFK equation (1.49). Following AFK Definitions 1.21–1.24, a pair (d, r) is *admissible* when

$$0 < r < \frac{d-1}{2}, \quad nr(d-r) = d^2 - 1$$

for an integer $n > 4$. Its associated real quadratic field is $K = \mathbb{Q}(\sqrt{n(n-4)})$. If $\epsilon > 1$ is the fundamental totally positive unit of K , $\Delta_0 = \text{disc}(K)$, and

$$f_j = \frac{\epsilon^j - \epsilon^{-j}}{\sqrt{\Delta_0}}, \quad r_{j,m} = \frac{f_{jm}}{f_j}, \quad d_{j,m} = r_{j,m+1} + r_{j,m}, \quad d_j = d_{j,1},$$

then (d, r) is equivalently indexed by (K, j, m) with $d = d_{j,m}$ and $r = r_{j,m}$. An admissible tuple

$$t = (d, r, Q) \sim (K, j, m, Q)$$

also contains a primitive integral indefinite form Q whose fundamental discriminant is Δ_0 and whose conductor f divides f_j [3, Definitions 1.27–1.28]. For $Q = \langle a, b, c \rangle$ we use the same symbol for the scaled Hessian $\begin{pmatrix} a & b/2 \\ b/2 & c \end{pmatrix}$, as AFK do, and write $\langle (p_1, p_2), (q_1, q_2) \rangle = p_2 q_1 - p_1 q_2$ and $\omega_d = e^{2\pi i/d}$.

Let A_t be AFK's signed generator of the level- d stability group of Q , let

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad L_{z,t} = \frac{d_j - 1}{2} I + \frac{f_j}{f} S Q,$$

and put

$$s_d(\mathbf{p}) = d + (1 + d)(1 + p_1)(1 + p_2).$$

For $\mathbf{p} \in \mathbb{Z}^2$, let $\mathcal{I}_{\mathbf{p}}$ be any complete set of representatives for $\mathbb{Z}^2/d\mathbb{Z}^2$ containing $\mathbf{0}$ and $\mathbf{p}$. AFK Definition 1.34 calls $\lambda \in \mathbb{Z}/d\mathbb{Z}$ a *shift* for t when $2\lambda + d_j - 1$ is coprime to d and, for every $\mathbf{p}$,

$$\sum_{\mathbf{q} \in \mathcal{I}_{\mathbf{p}}} \omega_d^{r(\mathbf{p}, (\lambda I + L_{z,t})\mathbf{q})} \text{shin}_{A_t}^{\mathbf{q}/d}(\rho_t) \text{shin}_{A_t^{-1}}^{(\mathbf{q}-\mathbf{p})/d}(\rho_t) = d^2 \delta_{\mathbf{p}, \mathbf{0}}^{(d)}. \quad (\text{TCC})$$

Here ρ_t is the positive root of Q , and the phases implicit in shin use the sign $(-1)^{s_d(\mathbf{p})}$ from AFK Definition 1.30. The set of shifts is denoted $\mathcal{Z}_t$. The *formal TCC*, AFK Conjecture 1.35, asserts $0, 1 \in \mathcal{Z}_t$ for every admissible tuple and equality of shift sets for admissible forms of the same discriminant. The twist attached to a shift satisfies

$$\det(G) f_t(\lambda) \equiv 1 \pmod{\bar{d}}, \quad f_t(\lambda) := r(2\lambda + d_j - 1 + d),$$

where $\bar{d} = d$ for odd d and $\bar{d} = 2d$ for even d .

Theorem 1 (Low-dimensional TCC). *For every admissible tuple in dimension four or five,*

$$0, 1 \in \mathcal{Z}_t.$$

If t, t' have the same dimension, then $\mathcal{Z}_t = \mathcal{Z}_{t'}$.

The dimension-four proof below is deliberately compact: it records the complete analytic and finite bridge but leaves the thirty-six explicit minor quotients to the supplementary certificate. Dimension five requires the longer powered-algebraicity and height-rigidity argument that forms the body of the paper.

2 Dimension four: the quadratic calibration

Put

$$\phi = \frac{1 + \sqrt{5}}{2}, \quad \beta_4 = \phi^2, \quad Q_4 = \langle 1, -3, 1 \rangle, \quad \tau_4 = -e^{\pi i/4}.$$

The complete signed reconstruction table is

$$T_4(x) = \begin{pmatrix} \sqrt{5} & -x & 1 & -x^{-1} \\ -x^{-1} & -x^{-1} & -x^{-1} & -x \\ 1 & -x^{-1} & 1 & x \\ -x & -x^{-1} & x & -x \end{pmatrix}, \quad (\text{D4.1})$$

where

$$x = \sqrt{2} \frac{S_2(\beta_4/4 \mid \beta_4, 1) S_2(1/4 \mid \beta_4, 1)}{S_2((\beta_4 + 1)/4 \mid \beta_4, 1)}. \quad (\text{D4.2})$$

The factor $\sqrt{2}$ is the reciprocal shift multiplier $2 \sin(\pi/4)$ obtained from (D4.FE) at $z = \beta_4/4$.

Lemma 2 (Dimension-four reconstruction table). *For the canonical tuple, the fifteen nonzero AFK analytic overlaps are the nonzero-characteristic entries of (D4.1). The entry $(T_4)_{0,0} = \sqrt{5}$ packages the normalized ghost identity coefficient into the same table.*

Proof. Put

$$\mathcal{D}_{c,b}^{(4)} := S_2\left(\frac{c + b\beta_4}{4} \mid \beta_4, 1\right).$$

For $(p, q) \neq (0, 0)$ and $r \equiv -p - q \pmod{4}$, the principal-form specialization of AFK Definition 1.32 is

$$\tilde{\nu}_{p,q} = (-1)^{4(p+q)+pq+\min(4,p+q)} \mathcal{D}_{4-p,q}^{(4)} \mathcal{D}_{4-r,p}^{(4)} \mathcal{D}_{4-q,r}^{(4)}. \quad (\text{D4.DS})$$

This formula already includes the AFK phase. The complete substitution audit is

(p, q)	r	$\tilde{\nu}_{p,q}$	(p, q)	r	$\tilde{\nu}_{p,q}$
$(0, 1)$	3	$-x$	$(0, 2)$	2	1
$(0, 3)$	1	$-x^{-1}$	$(1, 0)$	3	$-x^{-1}$
$(1, 1)$	2	$-x^{-1}$	$(1, 2)$	1	$-x^{-1}$
$(1, 3)$	0	$-x$	$(2, 0)$	2	1
$(2, 1)$	1	$-x^{-1}$	$(2, 2)$	0	1
$(2, 3)$	3	x	$(3, 0)$	1	$-x$
$(3, 1)$	0	$-x^{-1}$	$(3, 2)$	3	x
$(3, 3)$	2	$-x$			

Every nonzero-characteristic entry follows from (D4.DS) using only

$$S_2(z) S_2(\beta_4 + 1 - z) = 1, \quad S_2(z + 1) = 2 \sin(\pi z / \beta_4) S_2(z), \quad S_2(z + \beta_4) = 2 \sin(\pi z) S_2(z), \quad (\text{D4.FE})$$

and $\beta_4^2 = 3\beta_4 - 1$. The value $\sqrt{5}$ at $(0, 0)$ is not an analytic zero-characteristic cocycle value: after division by $\sqrt{5}$ it represents the Weyl coefficient $\mu_0 = 1$. $\square$

Define

$$K_{r,c}^{(4)}(x) = \frac{1}{4\sqrt{5}} \sum_{\substack{0 \leq a, b < 4 \\ r \equiv c+a \pmod{4}}} (T_4)_{a,b}(x) \tau_4^{ab} i^{bc}. \quad (\text{D4.3})$$

The representative reduction in (D4.3) retains the even-dimensional wrap sign. Its diagonal Fourier sum gives

$$\text{Tr } K^{(4)} = (T_4)_{0,0} / \sqrt{5} = 1. \quad (\text{D4.4})$$

Exact Laurent reduction proves that every one of its 36 rank-two minors belongs to the principal ideal

$$\left(x^2 - \sqrt{3 + \sqrt{5}}x + 1\right) \subset \mathbb{Q}(i, \sqrt{2}, \sqrt{5})[x, x^{-1}]. \quad (\text{D4.5})$$

The certificate records all 36 quotients coefficientwise in the genuine basis $1, \sqrt{2}, \sqrt{5}, \sqrt{10}$; two minors are structurally zero and the other 34 have nonzero formal quotients.

It remains to evaluate (D4.2). Write $K_4 = \mathbb{Q}(\sqrt{5})$, $\mathcal{O}_{K_4} = \mathbb{Z}[\phi]$, and label its embeddings by $\infty_1(\sqrt{5}) = +\sqrt{5}$ and $\infty_2(\sqrt{5}) = -\sqrt{5}$. Since 2 is inert, $|(\mathcal{O}_{K_4}/4)^\times| = 12$. The images of $\{\pm\phi^n\}$ together with the sign at ∞_2 have order 12; the ray exact sequence and $h_{K_4} = 1$ therefore give

$$\text{Cl}_{(4)\infty_2}(K_4) \cong C_2. \quad (\text{D4.6})$$

The field

$$L_4 = K_4(\sqrt{\phi})$$

has relative discriminant (4), and ∞_2 ramifies because $\phi' < 0$. The quadratic conductor–discriminant theorem and (D4.6) show that L_4 is the full ray field.

Set $u = \phi + \sqrt{\phi}$ and let $\sigma(\sqrt{\phi}) = -\sqrt{\phi}$. Then $u^\sigma = u^{-1}$ and

$$N_{L_4/K_4}(u) = 1, \quad \text{Tr}_{L_4/K_4}(u) = 1 + \sqrt{5}.$$

The field L_4 has signature $(2, 1)$ and discriminant -400 . For $s = \sqrt{\phi}$, the power-basis discriminant of $1, s, s^2, s^3$ is also -400 , equal to the certified field discriminant; hence $\mathcal{O}_{L_4} = \mathbb{Z}[s]$. Its Minkowski bound is $15/(2\pi) < 2.4$, and there is no ideal of norm two because

$$X^4 - X^2 - 1 \equiv (X^2 + X + 1)^2 \pmod{2}.$$

Thus $h_{L_4} = 1$. PARI/GP 2.15.4 [9] returns the fundamental units $s^3 - s = s^{-1}$ and $-s^2 + s = -u^{-1}$; its `bnfcertify=1` output certifies unconditionally that

$$R_{L_4} = \log \phi \log u.$$

For the nontrivial quadratic ray character χ , $L(s, \chi) = \zeta_{L_4}(s)/\zeta_{K_4}(s)$. Using

$$\zeta_F(s) = -\frac{h_F R_F}{w_F} s^{r_1+r_2-1} + O(s^{r_1+r_2})$$

for $F = K_4, L_4$ gives

$$L'(0, \chi) = \log u. \quad (\text{D4.7})$$

We now spell out the convention-sensitive specialization of Kopp's Theorem 1.1 [1]. Take modulus $4\mathcal{O}_{K_4}$, identity ray class $\mathfrak{A}$, $\mathfrak{b} = \mathcal{O}_{K_4}$, $\alpha = 4$, characteristic

$$\mathbf{r} = \begin{pmatrix} 0 \\ 1/4 \end{pmatrix}, \quad B_4 = \begin{pmatrix} 3 & -1 \\ 1 & 0 \end{pmatrix}^3 = \begin{pmatrix} 21 & -8 \\ 8 & -3 \end{pmatrix}.$$

Here $4(\beta_4\mathbb{Z} + \mathbb{Z}) = \mathfrak{bm}$, $4(r_2\beta_4 - r_1)\mathcal{O}_{K_4} = \beta_4\mathcal{O}_{K_4} = \mathcal{O}_{K_4} \in \mathfrak{A}$, and $r_2\beta'_4 - r_1 = \beta'_4/4 > 0$. Moreover $B_4\mathbf{r} - \mathbf{r} = (-2, -1)^\top$; B_4 is the positive generator of $\Gamma_{\mathbf{r}}$ with expanding eigenvalue β_4^3 . The fiber of

$$\text{Cl}_{(4)\infty_1\infty_2}(K_4) \longrightarrow \text{Cl}_{(4)\infty_2}(K_4)$$

has order two, so Kopp's exponent is $n = 2/2 = 1$.

Kopp defines a sign class $\mathfrak{R}$ in [1, Definition 6.3]. Here it is represented by $(-1 \bmod 4, + \text{ at } \infty_2)$. The residues of $\phi^n = a_n + b_n\phi$ for $0 \leq n < 6$ are

$$(a_n, b_n) = (1, 0), (0, 1), (1, 1), (1, 2), (2, 3), (3, 1),$$

and $\phi^6 \equiv 1 \pmod{4}$. Hence every unit congruent to -1 modulo 4 is $-\phi^{6k}$, which is negative at ∞_2 . Thus $\mathfrak{R} \neq \mathfrak{A}$; by (D4.6) it is the nonidentity class, and the differenced partial zeta is

$$Z_{(4)\infty_2}(s, \mathfrak{A}) = \zeta(s, \mathfrak{A}) - \zeta(s, \mathfrak{R}\mathfrak{A}) = L(s, \chi). \quad (\text{D4.8})$$

The Rademacher invariant of B_4 is zero: $s(21, 8) = -1/16$ gives $18/8 - 3 - 12(-1/16) = 0$. Thus $\psi^2(B_4) = 1$, while Kopp's theta character gives

$$\chi_{\mathbf{r}}(B_4) = \exp\left(2\pi i \frac{1}{2} \left(-\frac{28}{4} + \frac{152}{16}\right)\right) = i.$$

Consequently

$$(\psi^{-2}\chi_{\mathbf{r}}^{-1})(B_4) = -i = \Phi_{(0,1)}^2, \quad \Phi_{(0,1)} = e^{3\pi i/4}. \quad (\text{D4.9})$$

For completeness, put $C = \begin{pmatrix} 3 & -1 \\ 1 & 0 \end{pmatrix}$. The characteristic orbit in the three cocycle factors is

$$(0, 1) \mapsto (3, 0) \mapsto (1, 3) \mapsto (0, 1),$$

with unreduced steps

$$C(0, 1) = (-1, 0) = (3, 0) + (-4, 0), \quad C(3, 0) = (9, 3) = (1, 3) + (8, 0), \quad C(1, 3) = (0, 1).$$

Substitution in (D4.DS) gives exponent $4(0 + 1) + 0 + \min(4, 1) = 5$ and the three arguments $1 + \beta_4/4$, $1/4$, and $1 + (3\beta_4 - 1)/4$. Hence

$$\nu_{(0,1)} = \Phi_{(0,1)} \text{shin}_{B_4}^{\mathbf{r}}(\beta_4) = -x. \quad (\text{D4.10})$$

Squaring (D4.10) and applying Kopp's formula line by line now gives

$$\begin{aligned} x^2 &= \Phi_{(0,1)}^2 \text{shin}_{B_4}^{\mathbf{r}}(\beta_4)^2 \\ &= (\psi^{-2}\chi_{\mathbf{r}}^{-1})(B_4) \text{shin}_{B_4}^{\mathbf{r}}(\beta_4)^2 \\ &= \exp Z'_{(4)\infty_2}(0, \mathfrak{A}) = \exp L'(0, \chi) = u. \end{aligned} \quad (\text{D4.11})$$

Since $u + u^{-1} = 1 + \sqrt{5}$ and $x > 0$,

$$x + x^{-1} = \sqrt{3 + \sqrt{5}}.$$

Thus (D4.5) makes every minor vanish; (D4.4) then gives

$$\text{rank } K^{(4)} = 1, \quad (K^{(4)})^2 = K^{(4)}. \quad (\text{D4.12})$$

Lemma 3 (Dimension-four coefficient bridge). *For the canonical tuple $t_4 = (4, 1, \langle 1, -3, 1 \rangle)$, the equality $(K^{(4)})^2 = K^{(4)}$ is equivalent to the $\lambda = 1$ instance of AFK equation (1.49).*

Proof. For $\mathbf{p} \neq \mathbf{0}$, let $\nu_{\mathbf{p}}$ denote the analytic value from Lemma 2. Reciprocity gives

$$\nu_{\mathbf{p}}\nu_{-\mathbf{p}} = 1. \quad (\text{D4.13})$$

The actual ghost reconstruction is

$$K^{(4)} = \frac{1}{4}I + \frac{1}{4\sqrt{5}} \sum_{\mathbf{p} \neq \mathbf{0}} \nu_{\mathbf{p}} D_{\mathbf{p}}; \quad (\text{D4.14})$$

the artificial table entry $(T_4)_{0,0} = \sqrt{5}$ packages the first term and is not used as an analytic cocycle value.

Expand (D4.14) using $D_{\mathbf{p}}D_{\mathbf{q}} = \tau_4^{\langle \mathbf{p}, \mathbf{q} \rangle} D_{\mathbf{p}+\mathbf{q}}$. After multiplication by 80, the coefficient of $D_{\mathbf{p}}$ for $\mathbf{p} \neq \mathbf{0}$ is

$$\sum_{\substack{\mathbf{q} \bmod 4 \\ \mathbf{q} \neq \mathbf{0}, \mathbf{p}}} \tau_4^{\langle \mathbf{p}, \mathbf{q} \rangle} \nu_{\mathbf{p}-\mathbf{q}} \nu_{\mathbf{q}} - 2\sqrt{5} \nu_{\mathbf{p}}. \quad (\text{D4.15})$$

By (D4.13), the product is $\nu_{\mathbf{q}}/\nu_{\mathbf{q}-\mathbf{p}}$.

AFK's $\nu_{\mathbf{0}}^{\text{aux}}$ is a separate auxiliary cocycle value. Their zero-characteristic lemma gives

$$\nu_{\mathbf{0}}^{\text{aux}} + (\nu_{\mathbf{0}}^{\text{aux}})^{-1} = -(4-2)\sqrt{5} = -2\sqrt{5}. \quad (\text{D4.16})$$

Thus adjoining the omitted endpoints $\mathbf{q} = \mathbf{0}, \mathbf{p}$ to (D4.15) produces exactly the full endpoint-corrected sum

$$\sum_{\mathbf{q} \in \mathcal{I}_{\mathbf{p}}} \tau_4^{\langle \mathbf{p}, \mathbf{q} \rangle} \frac{\nu_{\mathbf{q}}}{\nu_{\mathbf{q}-\mathbf{p}}}, \quad (\text{D4.17})$$

with the endpoint ratios interpreted through $\nu_{\mathbf{0}}^{\text{aux}}$.

For nonzero characteristics write

$$\nu_{\mathbf{q}} = \Phi_{\mathbf{q}} \text{shin}_{A_{t_4}}^{\mathbf{q}/4}(\beta_4), \quad A_{t_4} = B_4.$$

The two phase identities in AFK Section 5.4 give

$$\begin{aligned} \frac{\Phi_{\mathbf{q}}}{\Phi_{\mathbf{q}-\mathbf{p}}} &= (-1)^{1+s_4(\mathbf{p})} \tau_4^{Q_4(\mathbf{p})} \tau_4^{(4-3)\langle \mathbf{p}, \mathbf{q} \rangle} \omega_4^{\langle \mathbf{p}, L_{z, t_4} \mathbf{q} \rangle}, \\ \frac{\text{shin}_{A_{t_4}}^{\mathbf{q}/4}(\beta_4)}{\text{shin}_{A_{t_4}}^{(\mathbf{q}-\mathbf{p})/4}(\beta_4)} &= \text{shin}_{A_{t_4}}^{\mathbf{q}/4}(\beta_4) \text{shin}_{A_{t_4}^{-1}}^{(\mathbf{q}-\mathbf{p})/4}(\beta_4). \end{aligned}$$

For the identity twist $f_{t_4}(1) = 1$. Its Weyl chirp contributes the additional $\tau_4^{\langle \mathbf{p}, \mathbf{q} \rangle}$, and hence

$$\tau_4^{(1+4-3)\langle \mathbf{p}, \mathbf{q} \rangle} = \omega_4^{\langle \mathbf{p}, \mathbf{q} \rangle}.$$

Up to a nonzero factor independent of $\mathbf{q}$, the coefficient equation is therefore

$$\sum_{\mathbf{q} \in \mathcal{I}_{\mathbf{p}}} \omega_4^{\langle \mathbf{p}, (I+L_{z, t_4}) \mathbf{q} \rangle} \text{shin}_{A_{t_4}}^{\mathbf{q}/4}(\beta_4) \text{shin}_{A_{t_4}^{-1}}^{(\mathbf{q}-\mathbf{p})/4}(\beta_4) = 16\delta_{\mathbf{p}, \mathbf{0}}^{(4)}. \quad (\text{D4.TCC})$$

This is exactly AFK equation (1.49) with $r = 1$ and $\lambda = 1$. For $\mathbf{p} = \mathbf{0}$, the cocycle inverse identity makes the sum 16 automatically. Every step is reversible, including the endpoint correction (D4.16). $\square$

By (D4.4) and (D4.12), $K^{(4)}$ is a trace-one rank-one idempotent, so Lemma 3 proves that 1 is a shift. The twist-shift congruence for the identity twist is

$$2\lambda + 7 \equiv 1 \pmod{8},$$

hence $\lambda \equiv 1 \pmod{4}$.

Complex conjugation replaces the twist G by $J_0 G$, where $J_0 = \text{diag}(1, -1)$. It preserves idempotence and sends

$$\lambda \longmapsto -(\lambda + d_j - 1) = 1 - \lambda \pmod{4}.$$

It therefore supplies the $\lambda = 0$ equation. Equivalently, $\det(J_0 G) = -1$ gives $-(2\lambda + 7) \equiv 1 \pmod{8}$ and hence $\lambda \equiv 0 \pmod{4}$.

Finally, dimension-four admissibility forces $r = j = m = 1$, $K_4 = \mathbb{Q}(\sqrt{5})$, and conductor one, so every admissible form has discriminant five. Class number one and the norm- -1 unit ϕ give narrow class number one; all such forms lie in one proper equivalence class. The AFK Section 7.1 transformation with $\mathbf{q} \mapsto \ell M \mathbf{q}$ gives $\mathcal{Z}_t \subseteq \mathcal{Z}_{t_M}$, and applying it to M^{-1} gives the reverse inclusion. Thus both shifts hold for every dimension-four admissible tuple and their shift sets are equal.

3 Dimension five: canonical data and conventions

Put

$$K = \mathbb{Q}(\sqrt{3}), \quad \beta = 2 + \sqrt{3}, \quad Q = \langle 1, -4, 1 \rangle, \quad L = \begin{pmatrix} 4 & -1 \\ 1 & 0 \end{pmatrix}.$$

Then

$$B = L^3 = \begin{pmatrix} 56 & -15 \\ 15 & -4 \end{pmatrix}, \quad B \begin{pmatrix} 0 \\ 1/5 \end{pmatrix} - \begin{pmatrix} 0 \\ 1/5 \end{pmatrix} = \begin{pmatrix} -3 \\ -1 \end{pmatrix}.$$

Label the real embeddings by $\infty_1(\sqrt{3}) = +\sqrt{3}$ and $\infty_2(\sqrt{3}) = -\sqrt{3}$.¹ The modulus throughout is

$$\mathfrak{m} = (5)\infty_2.$$

We continue to use the reciprocal double-sine convention (1). For $0 \leq c \leq 5$ and $0 \leq b < 5$, write

$$\mathcal{D}_{c,b} := S_2 \left(\frac{c + b\beta}{5} \middle| \beta, 1 \right).$$

For $(p, q) \neq (0, 0)$, let $r \equiv -p - q \pmod{5}$. The principal-form specialization of the AFK three-factor formula is

$$\tilde{\nu}_{p,q} = (-1)^{5(p+q)+pq+\min(5,p+q)} \mathcal{D}_{5-p,q} \mathcal{D}_{5-r,p} \mathcal{D}_{5-q,r}. \quad (2)$$

Formula (2) is used only for nonzero characteristics. Later we set the table entry $T_{0,0} := \sqrt{6}$ solely so that the identity Weyl coefficient is $T_{0,0}/\sqrt{6} = 1$. This reconstruction convention is not AFK's auxiliary zero-characteristic cocycle value.

Lemma 4 (Derivation and convention check for (2)). *Formula (2) is the specialization of AFK Definition 1.32 to $t = (5, 1, \langle 1, -4, 1 \rangle)$, with no unrecorded reciprocal or sign.*

Proof. AFK Definition 1.32 first forms $\tilde{\nu}_{\mathbf{p}} = \Phi_{\mathbf{p}} \text{shin}_B^{p/5}(\beta)$. Insert the defining modular-to-Jacobi quotient [3, equation (1.26)], and then use the Jacobi cocycle decomposition [3, equations (8.2)–(8.5)] and the S -kernel formula [3, equation (8.6)]. In the notation of AFK Section 8.2,

$$L = T^4 S, \quad B = L^3, \quad L\beta = \beta, \quad j_{L^k}(\beta) = \beta^k.$$

Thus the word $B = (T^4 S)^3$ contributes exactly three S -kernels. Modulo five the characteristic orbit is

$$(p, q) \mapsto (r, p) \mapsto (q, r) \mapsto (p, q), \quad r \equiv -p - q \pmod{5},$$

and the three unreduced linear arguments are

$$\frac{q\beta - p}{5}, \quad \frac{p\beta - r}{5}, \quad \frac{r\beta - q}{5}.$$

Reducing their first coordinates to the representatives $5-p, 5-r, 5-q$ gives respectively $\mathcal{D}_{5-p,q}$, $\mathcal{D}_{5-r,p}$, and $\mathcal{D}_{5-q,r}$. AFK's S -kernel is its displayed double sine in the denominator (AFK Section 8.2, equation for σ_S). That double sine is Kopp's $\text{Sin}_{2, \text{Kopp}}$; hence each denominator is precisely one factor of the reciprocal convention (1).

For completeness, apply the integral-translation law [3, equation (8.9)] to the three representative changes and multiply by the phase in AFK Definition 1.30. Direct substitution gives

$$\Phi_{p,q}(\text{three translation multipliers}) = (-1)^{E(p,q)}, \quad E(p,q) = 5(p+q) + pq + \min(5, p+q).$$

¹PARI orders real embeddings by increasing real root. Thus its first root of $y^2 - 3$ is $-\sqrt{3}$, so the PARI infinite-place vector $[1, 0]$ denotes the place called ∞_2 here. Equivalently, in the integral generator $\beta = 2 + \sqrt{3}$ used by the scripts, the first root of $y^2 - 4y + 1$ is $2 - \sqrt{3}$. Raw conductor outputs in the certificates use PARI's ordering.

This is a finite parity calculation:

$$E(p, q) \equiv \begin{cases} pq & (p + q \leq 5), \\ p + q + pq + 1 & (p + q \geq 5) \end{cases} \pmod{2},$$

where the two expressions agree at $p + q = 5$. In the order $p = 0, \dots, 4$ (columns $q = 0, \dots, 4$), the resulting signs are

$$\begin{pmatrix} * & + & + & + & + \\ + & - & + & - & + \\ + & + & + & + & - \\ + & - & + & + & + \\ + & + & - & + & - \end{pmatrix};$$

the asterisk is the excluded zero characteristic. This accounts for all twenty-four cases and proves (2) directly from the cited definitions. $\square$

4 Ray labels and all characteristics

Lemma 5 (Certified ray arithmetic). *The class and ray groups are*

$$\mathrm{Cl}(K) = 1, \quad \mathrm{Cl}_{(5)}(K) \cong C_4, \quad \mathrm{Cl}_{\mathfrak{m}}(K) \cong C_8, \quad \mathrm{Cl}_{(5)\infty_1\infty_2}(K) \cong C_8 \times C_2. \quad (3)$$

Kopp's residue/sign class $\mathfrak{R}$ is the element $4 \in C_8$. The fiber of $\mathrm{Cl}_{(5)\infty_1\infty_2}(K) \rightarrow \mathrm{Cl}_{\mathfrak{m}}(K)$ has order two, so the exponent in Kopp's theorem is $n = 2/2 = 1$. The computation is unconditional.

Proof. PARI/GP 2.15.4 gives the structures in (3), and `bnfcertify=1`. The unit group is $\{\pm\beta^n : n \in \mathbb{Z}\}$ and $\beta^3 \equiv 1 \pmod{5}$. Hence units congruent to -1 modulo 5 are $-\beta^{3k}$. Their value under ∞_2 is $-(2 - \sqrt{3})^{3k} < 0$, so no global unit realizes the pair $(-1 \bmod 5, + \text{ at } \infty_2)$ defining Kopp's $\mathfrak{R}$. It is therefore the nontrivial sign class, namely the unique element of order two in C_8 . $\square$

Proposition 6 (No quadratic-character shortcut). *The Kopp partial-zeta difference has zero coefficient at the unique order-two character of $\mathrm{Cl}_{\mathfrak{m}}(K)$. It is supported on four characters, all of order eight.*

Proof. Write $\mathrm{Cl}_{\mathfrak{m}}(K) = \langle g \rangle \cong C_8$ and $\mathfrak{R} = g^4$. For $\chi_k(g) = \exp(2\pi i k/8)$, Fourier inversion gives

$$\zeta_{\mathfrak{m}}(s, \mathfrak{A}) - \zeta_{\mathfrak{m}}(s, \mathfrak{R}\mathfrak{A}) = \frac{1}{8} \sum_{k=0}^7 \overline{\chi_k(\mathfrak{A})} (1 - \overline{\chi_k(\mathfrak{R})}) L(s, \chi_k). \quad (\text{CF})$$

Since $\chi_k(\mathfrak{R}) = (-1)^k$, the coefficient is zero for even k and two for odd k . The support is therefore $k = 1, 3, 5, 7$, whose characters all have order eight. The unique quadratic character is χ_4 ; it is even on $\mathfrak{R}$ and its coefficient in (CF) is zero. Hence the difference cannot factor through the quadratic quotient. This is the precise reason the dimension-four quadratic L -function reduction does not recur. Equivalently, $\mathfrak{R} = g^4$ lies in the kernel $\langle g^2 \rangle$ of the quotient $C_8 \rightarrow C_2$, so $\mathfrak{A}$ and $\mathfrak{R}\mathfrak{A}$ have the same image in the unique quadratic quotient and their difference projects to zero. $\square$

For each $(p, q) \neq (0, 0)$ start with p and repeatedly subtract five; let $\tilde{p}$ be the first lift for which

$$q(2 - \sqrt{3}) - \tilde{p} > 0. \quad (4)$$

Let $\mathfrak{q}$ be the prime above 3, and use it as the generator of C_8 . PARI's raw cyclic basis g gives the discrete logarithms of the positive principal ideals $(q\beta - \tilde{p})$ in the audit transcript, with

$\mathfrak{q} = g^5$. Multiplication by 5, which is its own inverse modulo 8, converts those raw logs to the normalized $\mathfrak{q}$ -coordinates below:

$$\begin{array}{c|ccccc}
 p \backslash q & 0 & 1 & 2 & 3 & 4 \\
 \hline
 0 & - & 0 & 6 & 2 & 4 \\
 1 & 4 & 7 & 5 & 7 & 0 \\
 2 & 2 & 5 & 5 & 6 & 3 \\
 3 & 6 & 7 & 2 & 1 & 1 \\
 4 & 0 & 4 & 3 & 1 & 3
 \end{array} \tag{5}$$

The corresponding positive lifts are 0 in the first row, -4 in row one except at $(1, 4)$ where it is 1, and $-3, -2, -1$ in rows two, three, and four.

Lemma 7 (All multiplier comparisons). *For every nonzero $(p, q) \bmod 5$, Kopp's modular multiplier equals the square of the AFK phase at that characteristic.*

Proof. Write $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\mathbf{r} = (\tilde{p}/5, q/5)^\top$, with $\tilde{p}$ chosen by (4). Substitution in Kopp's defining theta-character gives the exponent

$$\frac{1}{2} \{ (c - d + 1)r_1 + (-a + b + 1)r_2 - cdr_1^2 + 2(a - 1)dr_1r_2 - (a - 2)br_2^2 \}. \tag{6}$$

For $B = \begin{pmatrix} 56 & -15 \\ 15 & -4 \end{pmatrix}$, substituting the particular positive lift from (4) and reducing (6) modulo one gives

$$\frac{p^2 - 4pq + q^2}{5} \pmod{1} \tag{7}$$

separately for each of the twenty-four nonzero residue pairs (p, q) . Thus (7) is a certified finite case calculation with the stated lifts, not a polynomial identity for arbitrary integral p, q . Also $s(-4, 15) = -19/90$, so the Rademacher invariant is three and $\psi^2(B) = i$. Therefore the Kopp multiplier has exponent

$$-\frac{1}{4} - \frac{p^2 - 4pq + q^2}{5} \pmod{1}, \tag{8}$$

which is precisely the exponent of the square of the AFK phase. The exact twenty-four-row substitution, including every positive lift, is part of the bridge certificate. $\square$

For a normalized ray log k , name the positive square root of its ray value by

$$\begin{array}{c|ccccccc}
 k & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\
 \hline
 \text{positive root} & x & w & y^{-1} & z & x^{-1} & w^{-1} & y & z^{-1}.
 \end{array}$$

The class log in (5) selects the unsigned entry and the explicit sign in (2) selects its sign. Thus the complete reconstruction table is

$$T_5 = \begin{pmatrix} \sqrt{6} & x & y & y^{-1} & x^{-1} \\ x^{-1} & -z^{-1} & w^{-1} & -z^{-1} & x \\ y^{-1} & w^{-1} & w^{-1} & y & -z \\ y & -z^{-1} & y^{-1} & w & w \\ x & x^{-1} & -z & w & -z \end{pmatrix}. \tag{9}$$

Here x, y, z, w denote the four positive representatives of the ray-class pairs

$$(0, 4), \quad (6, 2), \quad (3, 7), \quad (1, 5), \tag{10}$$

respectively. In particular, the table is generated from the ray logs and the sign exponent; it is not stored as a list of expected answers in the bridge generator.

5 The analytic ray packet

Set

$$\begin{aligned} P(U) = & U^8 - (8 + 5\sqrt{3})U^7 + (53 + 30\sqrt{3})U^6 \\ & - (156 + 90\sqrt{3})U^5 + (225 + 130\sqrt{3})U^4 \\ & - (156 + 90\sqrt{3})U^3 + (53 + 30\sqrt{3})U^2 \\ & - (8 + 5\sqrt{3})U + 1. \end{aligned} \tag{11}$$

It is the reciprocal lift $P(U) = U^4 F(U + U^{-1})$ of

$$\begin{aligned} F(V) = & V^4 - (8 + 5\sqrt{3})V^3 + (49 + 30\sqrt{3})V^2 \\ & - (132 + 75\sqrt{3})V + (121 + 70\sqrt{3}). \end{aligned} \tag{12}$$

Kopp previously recognized the same polynomial numerically as the Stark-unit polynomial for $\mathbb{Q}(\sqrt{3})$ and modulus five; see [5, §7, (7.21)] and [6, Example 1.17, (3.12)]. PARI's `bnrstark` reproduces (12), but that recognition is not used as a proof below.

The square roots of the reciprocal packet have absolute polynomial

$$\begin{aligned} Q(X) = & X^{32} - 16X^{30} + 95X^{28} - 260X^{26} + 355X^{24} - 348X^{22} \\ & + 388X^{20} - 300X^{18} + 195X^{16} - 300X^{14} + 388X^{12} \\ & - 348X^{10} + 355X^8 - 260X^6 + 95X^4 - 16X^2 + 1. \end{aligned} \tag{13}$$

Define $w^{-1}, z^{-1}, x^{-1}, y^{-1}, y, x, z, w$ to be the unique positive roots of (13) in, respectively,

$$\begin{aligned} & (0.424835, 0.424836), (0.481476, 0.481477), (0.506963, 0.506964), (0.782390, 0.782391), \\ & (1.278133, 1.278134), (1.972526, 1.972527), (2.076946, 2.076947), (2.353849, 2.353850). \end{aligned} \tag{14}$$

Sturm's theorem certifies exactly one root in every interval.

5.1 Shintani's theorem with an explicit exponent

For a ray class $A \in \text{Cl}_{\mathfrak{m}}(K)$ put

$$X_A := \exp Z'_{\mathfrak{m}}(0, A).$$

Lemma 8 (Specialized Shintani algebraicity). *For every $A \in \text{Cl}_{\mathfrak{m}}(K)$,*

$$X_A^{5760} \in \mathcal{O}_H^{\times},$$

where H is the ray field of modulus $\mathfrak{m}$. These powers have the Artin action indexed by the ray classes. At every embedding of H inducing ∞_2 on K , their absolute value is one.

Proof. Work first in the full narrow ray group

$$C := \text{Cl}_{(5)\infty_1\infty_2}(K) \cong C_8 \times C_2.$$

In Shintani's notation, let μ be the class represented by $11 - 5\beta$, which is congruent to 1 modulo 5 and has signs $(-, +)$ at (∞_1, ∞_2) . Let ν be the class represented by 4, which is congruent to -1 modulo 5 and is totally positive. The exact ray logs are

$$\mu = [4, 1] \in C, \quad \nu = [4, 0] \in C.$$

Forgetting ∞_1 kills μ and sends ν to the unique order-two element $4 \in C_8$. Hence

$$G = \langle \mu \rangle, \quad \nu \notin G, \quad C/G \simeq \text{Cl}_{\mathfrak{m}}(K).$$

Every unit is $\pm\beta^n$, and β has order three modulo 5. Thus no totally positive unit is -1 modulo 5, which is Shintani's condition (0-3). There is no norm- -1 unit, since $a^2 - 3b^2 = -1$ is impossible modulo 3; this implies condition (0-6).

The field H has signature $(8, 4)$ and a unique octic subfield

$$M \simeq \mathbb{Q}(\zeta_{60})^+.$$

An exhaustive degree-eight subfield enumeration of the degree-sixteen defining field returns exactly one subfield; an exact `nfisom` calculation then identifies it with the displayed cyclotomic field. This is recorded in the archived `scripts/dimension_five_shintani_audit.gp` transcript and checked independently in `scripts/dimension_five_special_reduction_audit.gp`. Thus H/M is quadratic, M is the maximal absolutely abelian subfield of H , and exactly one real place of K splits in H . These are Shintani's hypothesis (0-9). His Theorem 2 [7] therefore applies to $X_{(5)}(c, G)$.

This invariant is exactly the one-place zeta difference used here. Indeed, if c maps to A , then the inverse images of A and RA in C are respectively

$$\{c, c\mu\}, \quad \{c\nu, c\mu\nu\}.$$

Shintani's Theorem 1 consequently gives

$$\begin{aligned} \log X_{(5)}(c, G) &= \zeta'_{\text{full}}(0, c) - \zeta'_{\text{full}}(0, c\nu) \\ &\quad + \zeta'_{\text{full}}(0, c\mu) - \zeta'_{\text{full}}(0, c\mu\nu) \\ &= \zeta'_m(0, A) - \zeta'_m(0, RA) = Z'_m(0, A). \end{aligned}$$

It remains to identify the imaginary quadratic field in Shintani's proof and to clear every exponent in his construction. Write $\mathcal{N}$ for the full two-infinite-place ray field, the normal closure of H . In the cyclic coordinates

$$C = \langle e_1 \rangle \times \langle e_2 \rangle \cong C_8 \times C_2$$

returned by the exact ray computation, the nontrivial automorphism $\sqrt{3} \mapsto -\sqrt{3}$ acts by

$$e_1 \mapsto (5, 0), \quad e_2 \mapsto (4, 1). \tag{S1}$$

Consequently its fixed subgroup is

$$C^{\text{fix}} = \{(a, b) : a + b \equiv 0 \pmod{2}\} = \langle (2, 0), (1, 1) \rangle. \tag{S2}$$

Here G_1 in Shintani's proof is trivial. His index-two subgroup $(C/G_1)_0$ has the property that $c^\sigma c^{-1} \in G_1$ for each of its classes, so it is contained in (S2); since both have index two, they are equal. Exact class-field construction with the HNF matrix $\begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$ gives

$$\mathcal{N}^{C^{\text{fix}}} = K(\sqrt{-5}). \tag{S3}$$

Shintani's field L is therefore the biquadratic field Kk with

$$k = \mathbb{Q}(\sqrt{-5}).$$

This derives the imaginary quadratic field occurring in his proof; it does not select one by numerical recognition from a list of quadratic subfields.

For an independent class-field check, the ray group of k modulo 15 is $C_{40} \times C_2$. Killing the subgroup with HNF matrix $\text{diag}(8, 2)$ produces a degree-sixteen extension of k . Its exact conductor, in the integral basis used by PARI, is

$$\mathfrak{c} = \begin{pmatrix} 15 & 0 \\ 0 & 3 \end{pmatrix}, \tag{S4}$$

and `nfisom` identifies its absolute defining field with $\mathcal{N}$. Thus it is the ray-class subfield over the very same k forced by (S1)–(S3). The computation also gives $h_k = 2$ and `bnfcertify`=1.

We now spell out the exponent bookkeeping. Factor

$$\mathfrak{c} = \mathfrak{p}_3 \bar{\mathfrak{p}}_3 \mathfrak{p}_5, \quad \mathfrak{p}_3 = \begin{pmatrix} 3 & 2 \\ 0 & 1 \end{pmatrix}, \quad \bar{\mathfrak{p}}_3 = \begin{pmatrix} 3 & 1 \\ 0 & 1 \end{pmatrix}, \quad \mathfrak{p}_5 = \begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix}.$$

For $\mathfrak{d} \mid \mathfrak{c}$, let $f_{\mathfrak{d}}$ be the smallest positive rational integer in $\mathfrak{d}$, let $h(\mathfrak{d})$ be the order of its ray group, and let $w(\mathfrak{d})$ be the number of roots of unity in k congruent to 1 modulo $\mathfrak{d}$. Here $h(\mathfrak{c}) = 16$. Put, exactly as in Shintani's equation (8),

$$n(\mathfrak{d}) = w(\mathfrak{d}) \frac{h(\mathfrak{c})}{h(\mathfrak{d})}. \quad (\text{S5})$$

The exact divisor audit is

$\mathfrak{d}$	$f_{\mathfrak{d}}$	$h(\mathfrak{d})$	$w(\mathfrak{d})$	$n(\mathfrak{d})$	clearing exponent
$\mathcal{O}_k$	1	2	2	16	$12h_k n(\mathfrak{d}) = 384$
$\mathfrak{p}_3$	3	2	1	8	$12f_{\mathfrak{d}} n(\mathfrak{d}) = 288$
$\bar{\mathfrak{p}}_3$	3	2	1	8	288
(3)	3	4	1	4	144
$\mathfrak{p}_5$	5	4	1	4	240
$\mathfrak{p}_3 \mathfrak{p}_5$	15	8	1	2	360
$\bar{\mathfrak{p}}_3 \mathfrak{p}_5$	15	8	1	2	360
$\mathfrak{c}$	15	16	1	1	180

(S6)

Indeed, Shintani's equations (4) and (9) write, for $\mathfrak{d} \neq \mathcal{O}_k$,

$$Z_{\mathfrak{d}} = |\varphi_{\mathfrak{d}}|^{1/(6f_{\mathfrak{d}})}, \quad W_{\mathfrak{c}} = \prod_{\mathfrak{d} \mid \mathfrak{c}} |Z_{\mathfrak{d}}(\cdots)|^{1/n(\mathfrak{d})}.$$

Replacing an absolute value by the product with its complex conjugate introduces the additional factor two, so $12f_{\mathfrak{d}} n(\mathfrak{d})$ clears every denominator. For $\mathfrak{d} = \mathcal{O}_k$, Shintani's equation (6) has exponent $1/(6h_k)$ instead, giving $12h_k n(\mathfrak{d})$. Proposition 4 is, literally in his notation,

$$Y_{(5)}(c, G) = \prod_{h \in H_1} \frac{W_{\mathfrak{c}}(\tilde{c} \nu h)}{W_{\mathfrak{c}}(\tilde{c} h)}. \quad (\text{S7})$$

Thus every W -invariant occurs with integral exponent $+1$ or -1 ; there is no additional rational exponent. The least common multiple of the last column of (S6) is

$$\text{lcm}(384, 288, 288, 144, 240, 360, 360, 180) = 5760. \quad (\text{S8})$$

Finally, (5) is prime and its only proper divisor fails Shintani's condition (0-3), so his $Y_{(5)}$ equals $X_{(5)}$ and no induction exponent is added. Proposition 5 now supplies the unit, Artin, and nonsplit-modulus-one assertions. $\square$

5.2 The labeled candidate field

The absolute polynomial of a root of (11) is

$$\begin{aligned} p(T) = & T^{16} - 16T^{15} + 95T^{14} - 260T^{13} + 355T^{12} - 348T^{11} \\ & + 388T^{10} - 300T^9 + 195T^8 - 300T^7 + 388T^6 - 348T^5 \\ & + 355T^4 - 260T^3 + 95T^2 - 16T + 1. \end{aligned}$$

Lemma 9 (Exact field and Frobenius labels). *The field defined by (11) is the labeled ray field H over K . Let U be the root of p in*

$$(3.890, 3.891).$$

Then U is a norm-one unit. Arithmetic Frobenius at the prime $\mathfrak{q}$ above 3 acts transitively on its eight real conjugates, which lie successively in

$$(3.890, 3.891), (5.540, 5.541), (0.612, 0.613), (4.313, 4.314), \\ (0.257, 0.258), (0.180, 0.181), (1.633, 1.634), (0.231, 0.232).$$

At the embeddings above ∞_2 , all conjugates of U have absolute value one.

Proof. Applying `rnfequation` to both the candidate and the ray field, `nfisisom` returns eight maps. Exact substitution shows that all eight maps fix the labeled root of $y^2 - 4y + 1$; hence the isomorphism is over K , not merely over $\mathbb{Q}$. Moreover `bnfcertify=1`, `bnfisunit` gives an exact unit decomposition, and the norm of U is one.

The prime $\mathfrak{q}$ generates the ray group. Modulo $\mathfrak{q}$ the relative polynomial is

$$T^8 + T^7 + 2T^6 + 2T^2 + T + 1 \in \mathbb{F}_3[T],$$

which is irreducible. Arithmetic Frobenius is therefore $T \mapsto T^3$. Reduction modulo this polynomial identifies one of the exact `nfgaloisconj` automorphisms with Frobenius.

The certificate refines $(3.890, 3.891)$ to a rational subinterval of width 10^{-80} containing exactly one root of p . Exact rational interval evaluation of the successive Frobenius polynomials places their images inside the eight disjoint intervals in the statement. This proves the labels Galois-equivariantly.

Finally, under the nonidentity embedding of K , substitute $T = 2(Y^2 - 1)/(Y^2 + 1)$ in the trace polynomial (12). The exact `nfpolsturm` counts are $[8, 0]$, so all four trace roots at that embedding lie in $(-2, 2)$. Each corresponding reciprocal quadratic has both roots on the unit circle. $\square$

5.3 Certified removal of the power

Theorem 10 (Unconditional ray packet). *For*

$$\mathcal{X}_k := \exp Z'_m(0, \mathfrak{q}^k), \quad 0 \leq k < 8,$$

the exact Artin-labeled packet is

$$(\mathcal{X}_0, \dots, \mathcal{X}_7) = (x^2, w^2, y^{-2}, z^2, x^{-2}, w^{-2}, y^2, z^{-2}). \quad (15)$$

In particular, $\mathcal{X}_0 = U$.

Proof. Kopp's proved Kronecker-limit theorem, with the convention and multiplier checked above, gives

$$\tilde{\nu}_{p,q}^2 = \exp Z'_m(0, \mathfrak{A}_{p,q}); \quad (16)$$

here $\mathfrak{A}_{p,q}$ is the normalized ray class in (5). The four positive generators at characteristics

$$(0, 1), \quad (0, 2), \quad (2, 4), \quad (3, 3)$$

have $\mathfrak{q}$ -coordinates 0, 6, 3, 1, respectively.

We evaluate the regularized integral for the reciprocal double sine (1) with Arb balls through `python-flint` [10]. Every Simpson panel uses an Arb enclosure of the fourth derivative. All endpoints, widths, tolerances, and acceptance comparisons are exact rationals or outward-rounded balls. Near zero, the positive series of $H_c(t) = \sinh(ct/2)/(ct/2)$ gives the rational bound

$$|f(t) - f(0)| \leq 4t^2 \quad (0 \leq t \leq 10^{-4}).$$

For the tail, every reduced argument and its complement belongs to $(1/100, 5)$; beyond 36, each exponential denominator exceeds $3/4$, giving the remainder bound $e^{-36}/9$. The candidate roots are selected by disjoint rational isolating intervals, independently of library ordering.

For each of the four characteristics, the Arb ball for $\log(\tilde{v}_{p,q}^2) - \log(U_k)$ contains zero. The largest upper bound for the absolute value of any of these four balls is

$$\epsilon = 4.4 \times 10^{-11}.$$

These four enclosures are logically needed before the packet has been identified: they cover one representative of each reciprocal pair of real ray classes, and their reciprocal partners bound the other four real embeddings. Once equality at one class is proved, the exact Artin action propagates it through the packet. Thus the extra three enclosures certify the eight-place height bound and the characteristic labels; they do not replace any step of Galois propagation.

By Lemma 8, both $\mathcal{X}_0^{5760}$ and U^{5760} are units in the same labeled field. Set

$$\eta = \mathcal{X}_0^{5760}/U^{5760}.$$

Put

$$\delta := 5760\epsilon < 2.54 \times 10^{-7}.$$

Lemma 9 and the four reciprocal-pair enclosures show that at every one of the eight real embeddings,

$$|\log |\sigma(\eta)|| < \delta.$$

At the four complex pairs, both numerator and denominator have absolute value one. Since H has signature $(8, 4)$ and degree sixteen, and since the finite-place terms vanish because η is a unit, the absolute logarithmic Weil height is therefore

$$\begin{aligned} h(\eta) &= \frac{1}{16} \left(\sum_{\sigma \text{ real}} \max(0, \log |\sigma\eta|) + 2 \sum_{\tau \text{ complex pair}} \max(0, \log |\tau\eta|) \right) \\ &< \frac{8}{16} \delta < 1.27 \times 10^{-7}. \end{aligned} \tag{H}$$

If $3 \leq d = [\mathbb{Q}(\eta) : \mathbb{Q}] \leq 16$ and η is not a root of unity, Voutier's explicit bound [8] gives

$$h(\eta) > \frac{1}{4d} \left(\frac{\log \log d}{\log d} \right)^3.$$

Its minimum for $3 \leq d \leq 16$ is $5.2279533222 \dots \times 10^{-5}$, a contradiction. A non-torsion quadratic unit has height at least $\frac{1}{2} \log((1 + \sqrt{5})/2)$, and a rational unit is ± 1 . Thus η is a root of unity. Since H has a real embedding, its only roots of unity are ± 1 ; positivity at the distinguished embedding gives $\eta = 1$. Therefore $\mathcal{X}_0^{5760} = U^{5760}$ and positivity yields $\mathcal{X}_0 = U$. The exact Artin action in Lemma 8, together with the certified Frobenius orbit in Lemma 9, proves the full packet (15). $\square$

Since every reciprocal double-sine factor in (2) is positive at the displayed arguments, equations (2), (5), and (15) now derive the signed table (9) unconditionally.

6 Exact finite cancellation

Let $\zeta_5 = e^{2\pi i/5}$ and $\tau = -e^{\pi i/5} = \zeta_5^3$. Define, without reference to software,

$$\Pi_{r,c}^{(5)} := \frac{1}{5\sqrt{6}} \sum_{\substack{0 \leq a, b < 5 \\ r \equiv c+a \pmod{5}}} T_{a,b} \tau^{ab} \zeta_5^{bc}, \quad 0 \leq r, c < 5, \tag{17}$$

where T is (9). Equivalently, $\Pi^{(5)} = \frac{1}{5} \sum_{a,b} (T_{a,b}/\sqrt{6}) D_{a,b}$ for $D_{a,b} e_c = \tau^{ab} \zeta_5^{bc} e_{c+a}$. In particular,

$$\text{Tr } \Pi^{(5)} = T_{0,0}/\sqrt{6} = 1. \quad (18)$$

Proposition 11 (Exact 100-minor certificate). *For the algebraic values selected by (13)–(14), all one hundred 2-minors of the matrix (17) vanish exactly. Hence $\text{rank } \Pi^{(5)} = 1$ and $(\Pi^{(5)})^2 = \Pi^{(5)}$.*

Proof. Over the coefficient field $\mathbb{Q}(\zeta_5, \sqrt{6})$, (13) has four degree-eight factors. Let w be the interval-defined root in (14). Exact rational interval arithmetic applied to the `nfgaloisconj` polynomials proves that `nfgaloisconj` substitutions are

$$\begin{array}{c|cccccccc} & w & w^{-1} & y & y^{-1} & x & x^{-1} & z & z^{-1} \\ \hline \text{conjugate index} & 2 & 8 & 5 & 4 & 10 & 6 & 16 & 12. \end{array} \quad (19)$$

with every image lying inside its corresponding interval in (14). Moreover, exact `nfisincl` identities express $\sqrt{5}$ and $\sqrt{6}$ as polynomials in w ; rational interval evaluation selects their positive signs. Among the four factors, factor four is the unique factor on which both of these subfield elements equal the positive $\sqrt{5}$ and positive $\sqrt{6}$. This certifies, rather than numerically assumes, that the interval-defined tuple is the tuple used in the factor-four calculation.

Substitution in (9) and (17), followed by exact reduction modulo the selected factor, makes all one hundred minors zero. As a factor selection check, factors one, two, and three leave respectively 70, 100, 100 nonzero minors. Factor one preserves the positive $\sqrt{5}$ embedding but reverses $\sqrt{6}$; thirty minors vanish for structural reasons common to those two embeddings, while the remaining seventy detect the wrong $\sqrt{6}$ sign. Factors two and three already reverse $\sqrt{5}$ and leave all one hundred minors nonzero. Thus the factor check is not circular. This is exact number-field arithmetic, not floating-point evaluation. The complete script and transcript are supplied with the manuscript.

The matrix is nonzero because (18) gives trace one. Thus its rank is exactly one. Any rank-one matrix M satisfies $M^2 = (\text{Tr } M)M$, proving idempotency. $\square$

7 Dimension-five twisted convolution

Lemma 12 (Weyl coefficient bridge). *The matrix $\Pi^{(5)}$ of (17) is the standard-basis Weyl reconstruction of the canonical AFK ghost for $t = (5, 1, \langle 1, -4, 1 \rangle)$. Moreover, $(\Pi^{(5)})^2 = \Pi^{(5)}$ is equivalent to the $\lambda = 1$ instance of AFK equation (1.49):*

$$\sum_{\mathbf{q} \in \mathcal{I}_{\mathbf{p}}} \zeta_5^{\langle \mathbf{p}, (I+L_{z,t})\mathbf{q} \rangle} \text{shin}_{A_t}^{\mathbf{q}/5}(\beta) \text{shin}_{A_t^{-1}}^{(\mathbf{q}-\mathbf{p})/5}(\beta) = 25\delta_{\mathbf{p}, \mathbf{0}}^{(5)}. \quad (20)$$

Proof. For $\mathbf{p} \neq \mathbf{0}$, put $\nu_{\mathbf{p}} := \tilde{\nu}_{\mathbf{p}}$ as given by (2). The first two unconditional parts of AFK's construction give

$$\nu_{\mathbf{p}} \nu_{-\mathbf{p}} = 1. \quad (21)$$

Theorem 10, followed by (2) and (5), identifies these twenty-four numbers with the nonzero entries of T . The actual ghost reconstruction is therefore

$$\Pi^{(5)} = \frac{1}{5} I + \frac{1}{5\sqrt{6}} \sum_{\mathbf{p} \neq \mathbf{0}} \nu_{\mathbf{p}} D_{\mathbf{p}}. \quad (22)$$

The artificial entry $T_{0,0} = \sqrt{6}$ packages the first term of (22) into the single entry formula (17); it is not an analytic zero-characteristic value.

Now expand $(\Pi^{(5)})^2 - \Pi^{(5)}$ using $D_{\mathbf{p}}D_{\mathbf{q}} = \tau^{\langle \mathbf{p}, \mathbf{q} \rangle} D_{\mathbf{p}+\mathbf{q}}$ and compare every Weyl coefficient. For $\mathbf{p} \neq \mathbf{0}$, multiplication by 150 makes its coefficient

$$\sum_{\substack{\mathbf{q} \bmod 5 \\ \mathbf{q} \neq \mathbf{0}, \mathbf{p}}} \tau^{\langle \mathbf{p}, \mathbf{q} \rangle} \nu_{\mathbf{p}-\mathbf{q}} \nu_{\mathbf{q}} - 3\sqrt{6} \nu_{\mathbf{p}}. \quad (23)$$

Using (21), the product in (23) is $\nu_{\mathbf{q}}/\nu_{\mathbf{q}-\mathbf{p}}$.

AFK's $\nu_{\mathbf{0}}^{\text{aux}}$ is an auxiliary cocycle value, not the identity coefficient in (22). Their zero-characteristic lemma specializes to

$$\nu_{\mathbf{0}}^{\text{aux}} + (\nu_{\mathbf{0}}^{\text{aux}})^{-1} = -(5-2)\sqrt{6} = -3\sqrt{6}. \quad (24)$$

After adjoining the two omitted endpoints $\mathbf{q} = \mathbf{0}, \mathbf{p}$, their combined contribution is

$$\nu_{\mathbf{p}}(\nu_{\mathbf{0}}^{\text{aux}} + (\nu_{\mathbf{0}}^{\text{aux}})^{-1}) = -3\sqrt{6} \nu_{\mathbf{p}}.$$

Thus (23) is exactly the full endpoint-corrected sum

$$\sum_{\mathbf{q} \in \mathcal{I}_{\mathbf{p}}} \tau^{\langle \mathbf{p}, \mathbf{q} \rangle} \frac{\nu_{\mathbf{q}}}{\nu_{\mathbf{q}-\mathbf{p}}}, \quad (25)$$

with the two endpoint ratios interpreted using $\nu_{\mathbf{0}}^{\text{aux}}$.

Write $\Phi_{\mathbf{q}}$ for the AFK phase. Their phase-ratio calculation specializes, with $r = f_j/f = 1$ and $d_j = d = 5$, to

$$\frac{\Phi_{\mathbf{q}}}{\Phi_{\mathbf{q}-\mathbf{p}}} = (-1)^{1+s_5(\mathbf{p})} \tau^{Q(\mathbf{p})} \tau^{\langle \mathbf{p}, \mathbf{q} \rangle} \zeta_5^{\langle \mathbf{p}, L_{z,t} \mathbf{q} \rangle}. \quad (26)$$

The first two factors are nonzero and independent of $\mathbf{q}$. Multiplying the remaining τ factor by the displacement chirp in (25), and using $\tau^2 = \zeta_5$, gives exactly

$$\zeta_5^{\langle \mathbf{p}, (I+L_{z,t}) \mathbf{q} \rangle}.$$

The cocycle inverse identity

$$\frac{\text{shin}_{A_t}^{\mathbf{q}/5}(\beta)}{\text{shin}_{A_t}^{(\mathbf{q}-\mathbf{p})/5}(\beta)} = \text{shin}_{A_t}^{\mathbf{q}/5}(\beta) \text{shin}_{A_t^{-1}}^{(\mathbf{q}-\mathbf{p})/5}(\beta) \quad (27)$$

gives the two factors printed in (20). Here $A_t = B$, $L_{z,t} = L$, and $f_t(1) = 2 + 9 \equiv 1 \pmod{5}$, so the identity twist produces exactly the chirp in (25). The remaining factor depends on $\mathbf{p}$ but is nonzero and independent of $\mathbf{q}$. Hence the vanishing of every nonzero Weyl coefficient is equivalent to (20) for $\mathbf{p} \neq \mathbf{0}$. For $\mathbf{p} = \mathbf{0}$, the cocycle inverse identity makes the sum 25 automatically. This is the specialization of AFK equations (5.78)–(5.98), including rather than suppressing their endpoint correction. $\square$

Theorem 13 (Complete dimension-five TCC). *For every dimension-five admissible tuple t ,*

$$0, 1 \in \mathcal{Z}_t.$$

Furthermore $\mathcal{Z}_{t'} = \mathcal{Z}_t$ for any two dimension-five admissible tuples t, t' .

Proof. Proposition 11 and Lemma 12 prove the $\lambda = 1$ identity for the canonical tuple. The twist condition is

$$\det(G)f_t(\lambda) \equiv 1 \pmod{5}, \quad f_t(\lambda) = 2\lambda + 9.$$

Thus $G = I$ gives $2\lambda + 9 \equiv 1 \pmod{5}$, or $\lambda \equiv 1 \pmod{5}$.

Complex conjugation preserves idempotency and sends λ to $\bar{\lambda} = -(\lambda + d - 1) = 1 - \lambda \pmod{5}$. It therefore sends the verified shift 1 to shift 0. Equivalently the conjugate twist has determinant -1 , and $-(2\lambda + 9) \equiv 1 \pmod{5}$ gives $\lambda \equiv 0 \pmod{5}$.

Finally, dimension-five admissibility forces $r = 1$, $K = \mathbb{Q}(\sqrt{3})$, and $j = m = 1$. Since the relevant conductor is one, every admissible form has discriminant twelve. The ordinary class number is one, so primitive indefinite forms of discriminant twelve constitute one $\mathrm{GL}_2(\mathbb{Z})$ -equivalence class. (The narrow class number is not needed: determinant- -1 transformations are allowed.) Thus every admissible form is integrally equivalent to the canonical one in the sense required here.

For clarity, the transport does not invoke AFK's conditional transformation theorem. If t_M is obtained from t by $M \in \mathrm{GL}_2(\mathbb{Z})$, then unconditionally

$$A_{t_M} = M^{-1}A_tM, \quad L_{z,t_M} = M^{-1}L_{z,t}M, \quad \beta_{t_M} = M^{-1}\beta_t.$$

Kopp's covariance formula sends each cocycle factor at $(t_M, \mathbf{q})$ to the corresponding factor at $(t, \ell M \mathbf{q})$, with $\ell = \mathrm{sgn}(j_{M^{-1}}(\beta_t))$; when $\det M = -1$ both factors are complex-conjugated, which does not change the real TCC sum. Reindexing

$$\mathbf{q}' = \ell M \mathbf{q}, \quad \mathcal{I}'_{\ell M \mathbf{p}} = \ell M \mathcal{I}_{\mathbf{p}},$$

and using symplectic covariance transforms the complete left side of (20) for $(t_M, \mathbf{p})$ into that for $(t, \ell M \mathbf{p})$. Hence $\mathcal{Z}_t \subseteq \mathcal{Z}_{t_M}$. Applying the same calculation to M^{-1} gives the reverse inclusion and therefore equality. $\square$

Proof of Theorem 1. The dimension-four assertion is proved in Section 2; the dimension-five assertion is Theorem 13. $\square$

8 Outlook

The proofs above establish the formal TCC in dimensions four and five. A companion paper [2] applies the same convention-audited bridge to the discriminant-32 dimension-seven stratum and to both dimension-eight strata, using Shintani height rigidity in dimension seven and a combination of linear CM reinduction and quadratic phase reduction in dimension eight. The discriminant-8 dimension-seven stratum is not claimed there. Dimension six remains exceptional: its finite rank-one certificate is exact, but one oriented primitive order-six special value is not proved by the inputs used here.

9 Scope and reproducibility

Theorem 13 proves the complete formal TCC in dimension five. Together with Section 2, it proves Theorem 1. It does not prove a dimension-five SIC, the TCC in arbitrary dimensions, or Zauner's conjecture. The new number-theoretic step is Theorem 10: it proves the previously conjectural dimension-five ray packet by combining Kopp's proved limit formula with the applicable case of Shintani's proved algebraicity theorem and an effective height argument.

All reported certificates were regenerated on Linux 6.8.0-124-generic (x86-64, four virtual AMD EPYC 9354P cores) with PARI/GP 2.15.4, Python 3.12.3, python-flint 0.9.0, and FLINT 3.6.0. Representative wall times and peak resident memory were:

audit	wall time	peak RSS
Shintani field and exponent	0.9 s	22 MB
dimension-five exact minors	1.1 s	22 MB
embedding/factor selection	0.3 s	23 MB
Arb double-sine enclosures	33.8 s	30 MB.

The complete regression suite takes approximately three minutes on that platform. The deterministic submission archive contains the files below and a self-verifying root manifest. The manifest is named `ARCHIVE_CONTENTS.sha256`. Its persistent DOI will be inserted after deposit; a repository commit alone is not a substitute for that archive.

The supplementary package contains:

- the dimension-four 36-minor, ray, multiplier, and PARI certificates;
- the dimension-five 25-characteristic bridge and exact C_8 Fourier-support audit;
- the PARI transcript with `bnfcertify=1`, positive ray-class logs, and Sturm root isolation;
- the exact interval/conjugate/factor embedding certificate, formal 100-minor Laurent certificate, and four-factor reduction;
- the Shintani sign-class, fixed-field, conductor, and divisor-by-divisor exponent-5760 audit;
- the labeled K -isomorphism, unit, Sturm, Frobenius-orbit, and rigorous Arb height-gap certificates.

SHA-256 digests for every archived proof artifact are recorded in the archive-root manifest

`ARCHIVE_CONTENTS.sha256`.

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